NSW INDEPENDENT TRIAL EXAMS – 2021
MATHEMATICS STANDARD YEAR 11 EXAM
MARKING GUIDELINES

Section I

Question	Answer	Assessed Outcome	Band
1	D	MS-S1.1, MS11-9	3
2	B	MS-S1.2, MS11-7	3
3	A	MS-M1.1, MS11-3	3
4	B	MS-F1.1, MS11-5	3
5	B	MS-F1.2, MS11-6	3
6	D	MS-F1.3, MS11-6	3
7	C	MS-M1.1, MS11-9	3
8	B	MS-M2, MS11-3	3
9	D	MS-F1.1, MS11-8	3
10	C	MS-M1.1, MS11-3	4
11	A	MS-A1, MS11-10	4
12	C	MS-F1.1, MS11-5	4
13	B	MS-S1.1, MS11-2	4
14	C	MS-M1.2, MS11-9	4
15	D	MS-S1.2, MS11-10	4
16	D	MS-A1, MS11-9	4
17	A	MS-S1.2, MS11-7	4
18	A	MS-M1.2, MS11-4	5
19	D	MS-F1.1, MS11-6	5
20	D	MS-S2, MS11-8	6

Section II

Part	Answer	Mark	Outcome Assessed	Band
21	$1898 \times 10^{24} = 1.898 \times 10^{3} \times 10^{24}$ $= 1.898 \times 10^{27}$ $x = 27$	1	MS-M1.1, MS11-3	3
22	Let $t = 3$, then $V = 12.5 + (2.5 \times 3)$ $= 20$ L When full the bucket will hold 40 L.	1 1	MS-A2, MS11-1	3
23	Let $M = 3 \times 12 = 36$ (months), $A = 20$ Then $D = \dfrac{(36 \times 20)}{150}$ **= 4.8 mL**	1 1	MS-A1, MS11-6	3
24(a)	**The mode number of medals won was 1**	1	MS-S1.2, MS11-3	3
(b)	**8 countries won more than 4 medals** Percentage is $\dfrac{8}{20} \times 100$ **= 40%**	1 1	MS-M1.1, MS11-3	3
(c)	**5 number summary: 0, 1, 3.5, 6, 14** **(1 mark for lower and upper quartile,** **1 mark for median, lower and highest scores)**	2	MS-S1,2, MS11-7	3/4
(d)	IQR $= 6 - 1$ $= 5$	1	MS-M1.2, MS11-7	3
25	* \$1131.50 – (36.50 × 25) = \$219 received for overtime * Number of extra hours worked **= \$219 ÷ (\$36.50 × 1.5)** **= \$219 ÷ \$54.75** **= 4**	1 1	MS-F1.2, MS11-5	3
26	**Surface area = $2\pi r^2$ where $r = 15$** $= 2 \times \pi \times 15^2$ **= 1414 cm^2 (nearest whole number)**	1 1	MS-M1.2, MS11-5	3
27	$S = \dfrac{150[(2 \times 2) + (150 - 1) \times 2]}{2}$ $= \dfrac{150(4 + 298)}{2}$ **= 22 650**	1 1	MS-A1, MS11-2	3/4

Part	Answer	Mark	Outcome Assessed	Band
28	Average daily intake in KJ $= \frac{1445}{17} \times 100$ $= 8500$ Average daily calorie intake $= 8500 \times 0.2$ $= 1700$	 1 1	MS-M1.3, MS11-3	3/4
29	Janet's fuel consumption on her trip was: $\frac{60}{8.57}$ $= 7$ L/100 km The chart shows that her make of car has an average fuel consumption of 6.4 L /100 km Percentage difference is an increase of $\frac{0.6}{6.4} \times 100\%$ $= 9.38\%$	1 1 1	MS-F1.3, MS11-9	4
30	$r = \sqrt{\frac{3V}{\pi}}$ $r^2 = \frac{3V}{\pi}$ $\pi r^2 = 3V$ $V = \frac{\pi r^2}{3}$	 1 1	MS-A1, MS11-1	4
31	Kyle would most likely have spun the spinner whose expected probabilities match those closest to the outcomes in the table. For spinner 1: Expected number of Red $= \frac{3}{6} \times 120$ $= 60$ Expected number of Blue $= \frac{2}{6} \times 120$ $= 40$ Expected number of Green $= 20$ For spinner 2: Each of the colours has the same expected number of: $\frac{2}{6} \times 120 = 40$ Spinner 1 gives the closer outcomes, so Kyle most likely spun spinner 1.	 1 1	MS-S2, MS11-10	

Part	Answer	Mark	Outcome Assessed	Band
32(a)	Using the formula $A = \frac{h}{2}(a+b)$ $270 = \frac{20}{2}(15 + AB)$ $15 + AB = 27$ $AB = 12$	1 1	MS-M1.2-MS11-4	3/4
(b)	The top surface has area 12 cm × 12 cm = **144** cm^2 The base surface has area 15 cm × 15 cm = **225** cm^2 The four side surfaces have a total area of 4 × 270 cm^2 = **1080** cm^2 **Surface area of solid = (144 + 225 + 1080) cm^2 = 1449 cm^2**	1 1	MS-S1.2, MS11-4	4
33(a)	$\mathbf{C = 425 + 140h}$	1	MS-A2, MS11-2	3
(b)	Let $50 + 125h = 425 + 140h$ Then $15h = 225$ **h = 15 hours.** **Both companies would charge the same hiring charge for 15 hours use of the crane.**	1 1	MS-A1, MS11-1	3/4
(c)	Let $h = 12 \times 24 = 288$ (hours) **Company A: C = \$650 + (125 × 288) = \$36 650** **Company B: C = \$425 + (140 × 288) = 40 745** **Company A is cheaper for 12 days of the crane hire.**	1 1	MS-A2, MS11-6	3/4
34(a)	**6 months**	1	MS-A2, MS11-2	3/4
(b)	The amount paid each month will be the gradient of the graph. Using (6, 300) and (0, 75), the gradient is $\frac{300-75}{6} = 37.5$ **So, Marcus pays \$37.50 per month for his phone.**	1	MS-A2, MS11-2	3/4
(c)	$C = 75 + 37.5n$ where n is the number of months **(1 mark for the equation of the form $y = mx + b$ 1 mark for correct substitution for m and b)**	2	MS-A2, MS11-10	3/4
(d)	**Let n = 9 in the equation in (c) Then $C = 75 + (37.5 \times 9)$ = \$412.50**	1	MS-A1, MS11-3	3/3
(e)	**Let C = 750 in the equation in (c) Then $750 = 75 + 37.5n$ $37.5n = 675$ $n = 18$ Marcus took 18 months to pay off his phone.**	1 1	MS-A1, MS11-3	3/4

Part	Answer	Mark	Outcome Assessed	Band
35(a)	**Taxable income is the income received less any allowable deductions.**	1	MS-F1.2, MS11-10	3
(b)	**37%**	1	MS-F1.2, MS11-10	3
(c)	Tax payable on \$125 000 = **\$19 822 + 0.37 × (125 000 – 87 000)** = \$19 822 + 0.37 × \$38 000 **= \$33 882**	1 1	MS-F1.2, MS11-10	4
(d)	**Since PAYE tax (tax paid) = \$36 250 and the tax payable is \$33 882, Demetre would receive a refund of \$36 250 – \$33 882** **= \$2368**	1 1	MS-F1.2, MS11-10	4
36(a)	$\mathbf{1\ GBP = \frac{2.5}{4} = 0.625\ EUR}$	1	MS-A2, MS11-2	3
(b)	**In 2020, 4 GBP = 7 EUR** **Exchange rate is 1 GBP = 1.75 EUR**	1 1	MS-A2, MS11-2	4
(c)	From (b), 1 EUR $= \frac{1}{1.75} = 0.57$ GBP **So, 1000 EUR = 570 GBP**	 1	MS-A2, MS11-2	4
(d)	**Since 1 GBP = 1.75 EUR in 2020,** **then 1.75 EUR = 1.89 AUD** **So, 1 AUD =** $\mathbf{\frac{1.75}{1.89}}$ **EUR** **= 0.93 EUR**	1 1	MS-A2, MS11-2	5
37(a)	O^+ is the most common blood type, **Probability (both patients have this type) = 0.4 × 0.4** **= 0.16** **= 16%**	1	MS-S2, MS11-8	3/4
(b)	Probability (only one patient has O^+ blood) = Pr (O^+, any other type) + Pr (any other type, O^+) = (0.4 × 0.6) + (0.6 × 0.4) **= 0.48** (48%)	1 1	MS-S2, MS11-8	4
(c)	**Pr(at least one patient has AB^- blood)** **= 1 – (both patients have any other blood type)** = 1 – (0.99 × 0.99) **= 1.99%**	1 1	MS-S2, MS11-8	4/5

Part	Answer	Mark	Outcome Assessed	Band
38(a)	Interest earned over 4 years is \$540 **i.e., \$135 per year.** **Annual rate** $= \frac{135}{3000} \times 100\%$ $= 4.5\%$	1 1	MS-A2, F1,1 MS11-5	4
(b)	$I = 135n$	1	MS-A2, MS11-5	4
(c)	Let $I = \$877.50$ in (b) Then $877.50 = 135n$ $n = 6.5$ **years**	1	MS-A1, MS11-5	4
(d)	After 4 years, the value of the investment is \$3000 + \$540 **= \$3540** **Over the next 2 years the interest earned will be:** **\$3540 × 0.035 × 2 = \$247.80** **Total interest earned over 6 years = \$540 + \$247.80** = \$787.80	1 1 1	MS-F1.1, MS11-5	5
39	The radius of the small semicircles is 1.5 cm The radius of the large semicircle is 4.5 cm The radius of the circle is 2.25 cm **Area of shaded sections =** **Area of large semicircle – [(area of circle) + (area of 1 small semicircle)]** $= (1/2 \times \pi \times 4.5^2) - [(\pi \times 2.25^2) + (1/2 \times \pi \times 1.5^2)]\ \text{cm}^2$ $= (31.81) - [(15.90) + (3.53)]\ \text{cm}^2$ $= 31.81 - 19.43$ **= 12.4 cm² (correct to one decimal place)**	1 1 1 1	MS-M1.2, MS11-4	5
40	Volume of tank $= \frac{\pi r^2 h}{3}$ where $r = 4$ and $h = 10$ $= \frac{\pi \times 4^2 \times 10}{3}$ $= 167.55\ \text{m}^3$ **Capacity of tank** $= 167.55 \times 1000$ **L** **= 167 550 L** **Volume of water in tank** $= \frac{\pi \times 2.4^2 \times 6}{3}$ $= 36.19\ \text{m}^3$ **= 36 190 L** **Amount of water needed to fill the tank is:** **(167 550 – 36 190) litres** **= 131 360 L**	1 1 1 1	MS-M1.1, 1.2, MS11-10	5/6

Part	Answer	Mark	Outcome Assessed	Band
41(a)	**Each sector angle is 360° ÷ 8 = 45°**	1	MS-M1.2, MS11-9	3
(b)	Area $= [2 \times \frac{45}{360} \times \{(\pi \times R^2) - (\pi \times \left(\frac{R}{2}\right)^2\}]$ $= \frac{1}{4} \times [\pi R^2 - \frac{\pi R^2}{4}]$ $= \frac{1}{4} \times [\frac{3\pi R^2}{4}]$ $= \frac{3\pi R^2}{16}$	1 1 1	MS-M1.2, MS11-4 MS-A1	6
(c)	Let $\frac{3\pi R^2}{16} = 60$ $R^2 = \frac{60 \times 16}{3\pi}$ $R = \sqrt{\frac{60 \times 16}{3\pi}}$ $= \sqrt{101.86}$ $= 10$	 1 1	MS-A1, MS11-10	5/6